ALTERNATIVE INVESTMENT RETURNS

Learning Outcome

- calculate and interpret alternative investment returns both before and after fees

Hedge funds often use complex strategies to achieve outsized returns with low correlation with the broader market. These more complex strategies require more sophisticated portfolio management tools and a larger range of skills, making them more expensive to run. Rather than paying a very high flat management fee, investors demand that some of the compensation is aligned to the performance delivered by the strategy in the form of a performance fee.

Apart from the performance fee, there are also other customized, complex compensation arrangements that seek to align manager and investor incentives. These structures are designed to reward investors for early involvement, larger investments, and/or longer lockup periods. Managers benefit from applying specialized knowledge and skills to achieve steadily rising returns in excess of a designated hurdle rate. These complex fee structures impact returns for different investors in the same fund, as well as returns before and after fees across various alternative investments.

Another factor that can lock in or magnify losses for hedge funds is investor redemptions. Redemptions frequently occur when a hedge fund is performing poorly. Redemptions may require the hedge fund manager to liquidate some positions and potentially receive particularly disadvantageous prices when forced to do so by redemption pressures, while also incurring transaction costs.

Funds sometimes charge a **redemption fee** to discourage redemption and to offset the transaction costs for remaining investors in the fund. A **notice period** is a period (typically 30–90 days) in advance that investors may be required to notify a fund of their intent to redeem some or all their investment. This notice period allows the fund manager an opportunity to liquidate a position in an orderly fashion without magnifying the losses. A **lockup period** is the minimum holding period before investors are allowed to make withdrawals or redeem shares from a fund. This provision is intended to allow the hedge fund manager the required time to implement and potentially realize a strategy's expected results. If the fund receives a drawdown request shortly after a new investment, the lockup period forces the investors who made the request to stay in the fund for a period of time rather than be allowed to immediately withdraw. In addition, funds sometimes impose a **gate**, a provision that when implemented limits or restricts redemptions for a period of time, usually at the discretion of the fund manager. Investors should be aware of their liquidity needs before investing in a fund with restrictive provisions.

A hedge fund's ability to demand a long lockup period while raising a significant amount of investment capital depends a great deal on the reputation of either the firm or the hedge fund manager. Funds of hedge funds may offer more redemption flexibility than is afforded to direct investors in hedge funds because of special redemption arrangements with the underlying hedge fund managers, the maintenance of added cash reserves, access to temporary bridge-loan financing, or the simple avoidance of less liquid hedge fund strategies.

Ideally, redemption terms should be designed to match the expected liquidity of the assets being invested in, but even with careful planning, an initial drawdown can turn into

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Alternative Investment Returns

Custom Fee Arrangements

Alternative investments often involve customized fee arrangements combining management and performance-based fees that vary based on the size, timing, and/or terms of investor participation in the investment over time. Examples include the following:

- **Fees based on liquidity terms and asset size:** Limited partnerships may charge different rates depending on the liquidity terms that an investor is willing to accept (longer lockups resulting in lower fees), and managers may discount their fees for larger investors or for placement agents who introduced these investors. Different investors in the same fund may face different fee structures. For example, management fees for large LPs might range from 0.5% to 1.5%, with incentive fees reduced to 10%–15%, depending on the mandate. Such reductions can be meaningful in terms of net realized returns. However, smaller funds with strong performance (and capacity constraints) are able to maintain higher fees and may even decline business from larger investors rather than agree to a lower fee.
- **Founders shares:** As a way to entice early participation in startup funds, managers sometimes offer incentives known as **founders class shares**. Founders shares entitle investors to a lower fee structure, such as a 1.5% management fee and a 10% performance fee rather than a more standard 2% management fee and a 20% performance fee, and may apply only to the first $100 million in assets invested, although cutoff thresholds vary. Another alternative is to reduce the fees for early founders share investors once the fund achieves a critical mass or performance targets.
- **"Either/or" fees:** Major institutional investors, such as the Teacher Retirement System of Texas, have demanded that alternative investment funds accept an **either/or fee** agreement by choosing *between* fixed management and variable performance fees. Managers agree *either* to charge a lower, 1% management fee (to cover expenses during down years) *or* to accept a higher, 30% incentive fee above a mutually agreed-upon annual hurdle (to incentivize and reward managers during up years), whichever is greater. Major investors offering larger commitments may negotiate such novel fee structures designed to reward returns in excess of a benchmark, while smaller investors, such as high-net-worth individuals with smaller commitments, usually face more traditional fees.

Alternative Investment Return Calculations

Return calculations vary among alternative investments based on the form of the investments, as described in an earlier lesson. While more liquid alternative investments, such as REITs, commodity index exchange-traded funds, or other frequently traded investments, have a simple management fee structure similar to common assets, those with longer life cycles, illiquidity, and less transparency, such as private equity, hedge funds, and real estate, often use performance fees with modifications to create incentives for managers to act in the best interest of investors.

The impact of different fee arrangements and their effect on the resulting returns to investors is best illustrated using a series of examples. If we assume fixed GP management fees as a percentage of assets under management (AUM) of r_m, beginning-of-period assets of P_0, end-of-period assets of P_1, and a GP performance fee (p) that is a percentage of total return, the GP's return in currency terms (R_{GP}) is as follows:

$$R_{GP} = (P_1 \times r_m) + \max[0, (P_1 - P_0) \times p]. \quad 4$$

And we may solve for an investor's periodic rate of return, r_i, as follows:

$$r_i = (P_1 - P_0 - R_{GP})/P_0. \quad 5$$

EXAMPLE 3

Kettleside is a timberland investment management organization with $100 million of initial investment capital. It charges a 1% management fee based on year-end AUM (r_m) and a 20% performance fee (p). In its first year, Kettleside generates a 30% return. Assume management fees are calculated using an end-of-period valuation.

1. What are Kettleside's fees if the performance and management fees are calculated independently? What is an investor's effective return given this fee structure?

   Hide Solution

   Solution:

   Using Equation 4 with P_0 equal to $100 million, P_1 of $130 million, r_m = 1%, and p = 20%,

   R_{GP} = $130 million × 1% + max[0, ($130 million – $100 million) × 20%]

   = $7.3 million.

   r_i = ($130 million – $100 million – $7.3 million)/$100 million

   = 22.7%.

2. What are Kettleside's fees if the performance fee is calculated from the return *net* of the management fee? What is an investor's net return given this fee structure?

   Hide Solution

   Solution:

   In order to solve for GP return where performance fee is calculated *net* of management fees, we must modify Equation 4 as follows:

   $$R_{GP(Net)} = (P_1 \times r_m) + \max\{0, [P_1(1 - r_m) - P_0] \times p\}. \qquad 6$$

   Using Equation 6 with P_0 of $100 million, P_1 of $130 million, r_m = 1%, and p = 20%,

   $R_{GP(Net)}$ = $130 million × 1% + max{0, [$130 million(0.99) – $100 million] × 20%}

   = $7.04 million.

   r_i = ($130 million – $100 million – $7.04 million)/$100 million

   = 22.96%.

   Note: P_1 is equal to $130 million – $7.04 million = $122.96 million

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The previous example demonstrates how fee calculations affect investor returns, with a higher return when fees are calculated on a net basis. Alternative investment databases and indexes usually report fund performance net of aggregated fees. If fee structures vary, the actual net-of-fee returns earned by various investors may vary from the quoted return.

One fee structure variation involves the modification of performance fees using hurdle rates and high-water marks. In the next example, we consider the effect of both on investor returns using the same details from the prior example.

EXAMPLE 4

Kettleside Timberland LP Performance Fee Modifications

performance fee (p).

1. If Kettleside's fee agreement specifies a 6% hurdle rate with performance fees based on returns *in excess of* the hurdle rate, what are Kettleside's fees assuming the performance fee is calculated *net* of the management fee? What is an investor's net return given this fee structure?

Hide Solution

Solution:

To solve for Kettleside's fees with a hard hurdle rate calculated net of management fees, we make a further adjustment to Equation 4 by incorporating the hurdle rate, r_h, as follows:

$$R_{GP(Net\ with\ Hurdle)} = (P_1 \times r_m) + \max\{0, [P_1(1 - r_m) - P_0 \times (1 + r_h)] \times p\} \qquad 7$$

$R_{GP(Net\ with\ Hurdle)}$ = \$130 million × 1% + max{0, [\$130 million(0.99) – \$100 million(1.06)] × 20%}

= \$5.84 million.

r_i = (\$130 million – \$100 million – \$5.84 million)/\$100 million

= 24.16%.

2. In the second year, Kettleside fund value declines to \$110 million. The fee structure is as specified in Question 2 of Example 3 but also includes the use of a high-water mark (P_{HWM}) computed net of fees (note: no hurdle rate). What are Kettleside's fees in the second year? What is an investor's net return for the second year given this fee structure?

Hide Solution

Solution:

We must again alter Equation 4 to include the high-water mark (P_{HWM}) provision, as follows:

$$R_{GP(High\text{-}Water\ Mark)} = (P_2 \times r_m) + \max[0, (P_2\,(1 - r_m) - P_{HWM}) \times p], \qquad 8$$

where P_{HWM} is defined as the maximum fund value at the end of any *previous* period net of fees. We may solve for investor return r_i in Period 2 as follows:

$$r_i = (P_2 - P_1 - R_{GP})/P_1. \qquad 9$$

$R_{GP(High\text{-}Water\ Mark)}$ = \$110 million × 1% + max[0, (\$110 million – \$122.7 million) × 20%]

= \$1.1 million.

r_i = (\$110 million – \$122.96 million – \$1.1 million)/\$122.96 million

= –11.435%.

The beginning capital position in the second year for the investors is \$130 million − \$7.04 million = \$122.96 million. The ending capital position at the end of the second year is \$110 million − \$1.1 million = \$108.9 million.

3. In the third year, Kettleside's fund value increases to \$128 million. The fee structure is as specified in Questions 1 and 2 of Example 4. What are Kettleside's fees in the third year? What is an investor's net return for the third year given this fee structure?

Hide Solution

Solution:

$R_{GP(High\text{-}Water\ Mark)} = (P_3 \times r_m) + \max[0, (P_3\,(1 - r_m) - P_{HWM}) \times p]$.

$r_i = (P_3 - P_2 - R_{GP})/P_2$.

Note that the high-water mark, P_{HWM}, is the highest value of the fund after fees in all previous years. In Kettleside's case, it was $122.96 million, the ending value in the first year, P_1.

Kettleside Timberland LP Performance Fee Modifications

Year	Fund Value ($m), after Fees
0	100.00
1	122.70
2	108.90

High-Water

$R_{GP(High\text{-}Water\ Mark)}$ = $128 million × 1% + max[0, ($128 million × 0.99 – $122.96 million) × 20%]

= $2.032 million.

r_i = ($128 million – $2.032 million – $1.79 million)/$108.9 million

= 15.67%.

The beginning capital position in the third year for the investors is $110 million − $1.1 million = $108.9 million. The ending capital position for the third year is $128 million – $1.79 million = $126.21 million, which represents a new high-water mark to be applied the following year for this investor.

Performance fee modifications may have similar or different effects on the periodic investor returns depending on the timing of an investment. For example, if two Kettleside investors were to purchase the fund at different times and had otherwise similar fee structures, they would both realize a fee reduction in the case of a hard hurdle equal to $P_t \times r_h \times p$, or the product of the end-of-period fund value for year t, the hurdle rate, and the performance fee. However, in the case of a high-water mark, the time-dependent nature of this fee modification gives different results for an investor who enters the fund at a later date, as in the following example.

EXAMPLE 5

Kettleside Timberland LP High-Water Mark for New Investor

At the end of Year 2, Kettleside Timberland LP has capital of $108.9 million. Consider the Year 3 returns of a *new* investor assuming the same fund performance and fee structure—namely, a 1% management fee based on year-end AUM (r_m), a 20% performance fee (p), and a high-water mark provision.

$R_{GP(High\text{-}Water\ Mark)} = (P_3 \times r_m) + \max[0, (P_3\,(1 - r_m) - P_{HWM}) \times p]$.

$r_i = (P_3 - P_2 - R_{GP})/P_2$.

$R_{GP(High\text{-}Water\ Mark)}$ = $128 million × 1% + max[0, ($128 million × 0.99

– $108.9 million) × 20%]

= $4.85 million.

r_i = ($128 million – $108.9 million – $4.85 million)/$108.9 million

= 13.09.

since the fund's inception.

In other instances, the timing of returns can have a meaningful impact on manager fees and investor returns, as shown in the following example of a clawback provision.

EXAMPLE 6

Tenderledge Opportunity Fund LP—Clawback Provision

Tenderledge Opportunity Fund makes $20 million in new investments, evenly divided with $10 million into Arguston Inc. (a leveraged buyout) and $10 million to Heartfield Digital (an early-stage venture) One year later, Arguston is sold to a strategic buyer for $22 million after costs. Two years later, Heartfield Digital fails and Tenderledge is unable to recoup any of its original investment.

1. If Tenderledge's fee agreement as general partner (GP) specifies a 20% performance fee of aggregate profits (p) with a clawback provision, which performance fees will Tenderledge accrue and what will it ultimately receive?

   Hide Solution

   Solution:

   Gain in the Arguston investment: $22 million – $10 million = $12 million.

   Loss in the Heartfield Digital investment: $0 – $10 million = –$10 million.

   Aggregate gain of Tenderledge after two years = $12 million – $10 million = $2 million.

   Tenderledge would initially accrue 20% of the $12 million aggregate profit from the sale of Arguston at the end of the first year, or $12 million × 20% = $2.4 million. This amount is typically held in escrow for the benefit of the GP but not paid.

   The failure of Heartfield Digital in Year 2 reduces the original $12 million gain by $10 million, so the aggregate fund gain at the end of Year 2 is only $2 million. This net profit results in a performance fee of $400,000 (= $2 million × 20%). Tenderledge would then have to return $2 million of the previously accrued performance fees to LP investor capital accounts due to the clawback provision.

Knowledge Check Fee and Return Calculations

AWJ Capital is a hedge fund with $100 million of initial investment capital. It charges a 2% management fee based on year-end AUM and a 20% incentive fee. In its first year, AWJ Capital has a 30% return. Assume management fees are calculated using end-of-period valuation.

1. What are the fees earned by AWJ if the incentive and management fees are calculated independently? What is an investor's effective return given this fee structure?

   Hide Solution

   Solution:

   AWJ fees:

Total fees to AWJ Capital = $8.6 million.

2. What are the fees earned by AWJ assuming that the incentive fee is calculated from the return net of the management fee? What is an investor's net return given this fee structure?

Hide Solution

Solution:

$130 million × 2% = $2.6 million management fee.

($130 million − $100 million − $2.6 million) × 20% = $5.48 million incentive fee.

Total fees to AWJ Capital = $8.08 million.

Investor return = ($130 million − $100 million − $8.08 million)/$100 million = 21.92%.

3. If the fee structure specifies a hurdle rate of 5% and the incentive fee is based on returns in excess of the hurdle rate, what are the fees earned by AWJ assuming the performance fee is calculated net of the management fee? What is an investor's net return given this fee structure?

Hide Solution

Solution:

$130 million × 2% = $2.6 million management fee.

($130 million − $100 million − $5 million − $2.6 million) × 20% = $4.48 million incentive fee.

Total fees to AWJ Capital = $7.08 million.

Investor return = ($130 million − $100 million − $7.08 million)/$100 million = 22.92%.

4. In the second year, the fund value declines to $110 million. The fee structure is as specified for Question 1 but also includes the use of a high-water mark (computed net of fees). What are the fees earned by AWJ in the second year? What is an investor's net return for the second year given this fee structure?

Hide Solution

Solution:

$110 million × 2% = $2.2 million management fee.

No incentive fee because the fund has declined in value.

Total fees to AWJ Capital = $2.2 million.

Investor return = ($110 million − $2.2 million − $121.4 million)/$121.4 million = –11.20%.

The beginning capital position in the second year for the investors is $130 million − $8.6 million = $121.4 million. The ending capital position at the end of the second year is $110 million − $2.2 million = $107.8 million.

5. In the third year, the fund value increases to $128 million. The fee structure is as specified in Questions 1 and 4. What are the fees earned by AWJ in the third year? What is an investor's net return for the third year given this fee structure?

Hide Solution

Solution:

The $121.4 million represents the high-water mark established at the end of Year 1.

Total fees to AWJ Capital = $3.88 million.

Investor return = ($128 million − $3.88 million − $107.8 million)/$107.8 million = 15.14%. The ending capital position at the end of Year 3 is $124.12 million. This amount is the new high-water mark.

Relative Alternative Investment Returns and Survivorship Bias

Investors seeking higher risk-adjusted returns with low correlation with common asset classes in alternative investments often track their performance based on *relative* returns. As is the case for more common asset classes, returns on individual alternative investments are usually compared to a benchmark of investments with similar features. These benchmarks may be interpreted differently or take on different characteristics in the case of alternative investments. For example, the use of a composite benchmark for private equity or real estate investments may be misleading if a specific investment is in a different life cycle phase than most of its peers. However, return comparisons between such investments of the same vintage year on an annual or "since inception" basis lead to more accurate results. That said, lockups and illiquidity may prevent an investor from reacting to underperformance by selling an investment.

Hedge fund indexes deserve greater scrutiny given changes to the universe of funds included in a benchmark over time. For example, studies suggest that over a quarter of all hedge funds fail within the first three years due to performance problems that lead to investor defections and fund closure. The exclusion of failed funds from a given benchmark is a form of selection bias that can lead investors to overly optimistic return expectations known as **survivorship bias**. Survivorship bias is a major problem among hedge fund indexes that include only current investment funds and exclude those funds that are no longer available. A second form of bias relates to how and when hedge fund returns are initially included in a benchmark index. For example, a fund manager may launch several hedge fund investments at once and include only the most successful funds in an index a couple of years after inception. The subsequent inclusion or "backfilling" of prior performance data on a selective basis serves to increase average reported returns in what is known as **backfill bias**. Because of survivorship and backfill biases, hedge fund indexes may not reflect actual average hedge fund performance but, rather, only the returns of those hedge funds that initially performed best and/or have not failed.

EXAMPLE 7

Comparison of Returns: Investment Directly into a Hedge Fund or through a Fund of Hedge Funds

An investor is contemplating investing €100 million in either the ABC Hedge Fund (ABC HF) or the XYZ Fund of Funds (XYZ FOF). XYZ FOF has a "1 and 10" fee structure and invests 10% of its AUM in ABC HF. ABC HF has a standard "2 and 20" fee structure with no hurdle rate. Management fees are calculated on an annual basis on AUM at the beginning of the year. For simplicity, assume that management fees and incentive fees are calculated independently. ABC HF has a 20% return for the year before management and incentive fees.

1. Calculate the return to the investor from investing directly in ABC HF.

Hide Solution

Solution:

2%), and the incentive fee is €4 million (= €20 million × 20%). The return to the investor is 14% [= (20 − 2 − 4)/100].

2. Calculate the return to the investor from investing in XYZ FOF. Assume that the other investments in the XYZ FOF portfolio generate the same return before management fees as those of ABC HF.

Hide Solution

Solution:

XYZ FOF earns a 14% return, or €14 million profit after fees on €100 million invested with hedge funds. XYZ FOF charges the investor a management fee of €1 million (= €100 million × 1%) and an incentive fee of €1.4 million (= €14 million × 10%). The return to the investor is 11.6% [= (14 − 1 − 1.4)/100].

3. Why would the investor choose to invest in a fund of funds instead of a hedge fund given the effect of the "double fee" demonstrated in the answers to Questions 1 and 2?

Hide Solution

Solution:

This scenario assumes that returns are the same for all underlying hedge funds. In practice, this result will not likely be the case, and XYZ FOF may provide due diligence expertise and potentially valuable diversification. In addition, the underlying hedge fund might be closed to new investors and investing in the FOF may be the only way to access the hedge fund.

Question Set

1. Soft Hurdle

A real estate investment fund has deployed $100 million initial capital to purchase a property. The fund has a soft hurdle preferred return to investors of 8% per annum and an 80%/20% carried interest incentive split thereafter (with a standard catch-up clause). At the end of Year 2, the property is sold for a total of $160 million.

Ignoring management fees, what are the correct distributions to the LPs and the GP?

Hide Solution

Solution:

With a soft hurdle arrangement, the carried interest is calculated on the entire annual gross return as long as the set hurdle is exceeded. To calculate the distributions of gain, one needs to construct a waterfall of cash flows.

First, the LPs would be due their $100 million initial investment.

Then, they would be due $16 million (8% preferred return on initial capital for two years).

The soft hurdle has been met, and the GP is due the carried interest until 20% of the profits generated is received, or $4 million (2% for two years), which would be paid to the GP next as a catch-up to the achieved hurdle return.

The residual amount would be $160 million − $100 million − $16 million − $4 million = $40 million. This amount would then be split 80% to the LPs and 20%

So, the total payout with a soft annual hurdle of 8% of the $100 million would end up with the following waterfall:

	LP	GP
Return of Capital	$100 m	
8% Preferred per Annum	$16 m	
GP Catch-Up 20%		$4 m
80%/20% Split	$32 m	$8 m
Total Payout	$148 m	$12 m

2. Hard Hurdle

Following Question 1, how would the distributions to the LPs and GP have been different if the real estate investment fund had a hard hurdle of 8% per annum and no catch-up clause?

Hide Solution

Solution:

If the fund had a hard hurdle rate instead (i.e., no catch-up clause), only the amount above the $100 return of capital and $16 million preferred return would be subject to the 20% carried interest incentive to the GP: 20% × $44 million = $8.8 million—quite a bit less than the carried interest payment with the soft hurdle. The LPs would be due the balance of $35.2 million (= $44 million − $8.8 million incentive). This would result in the following total payout:

	LP	GP
Return of Capital	$100 m	
8% Preferred per Annum	$16 m	
80%/20% Split above Hurdle	$35.2 m	$8.8 m
Total Payout	$151.2 m	8.8 m

3. Calculating Net Return

Capricorn Fund of Funds invests GBP100 million in each of Alpha Hedge Fund and ABC Hedge Fund. Capricorn Fund of Funds has a "1 and 10" fee structure. Management fees and incentive fees are calculated independently at the end of each year. After one year, net of their respective management and incentive fees, Capricorn's investment in Alpha is valued at GBP80 million and Capricorn's investment in ABC is valued at GBP140 million. The annual return to an investor in Capricorn Fund of Funds, net of fees assessed at the fund-of-funds level, is closest to:

A. 7.9%.
B. 8.0%.
C. 8.1%.

Hide Solution

Solution:

A is correct, because the net investor return is 7.9%, calculated as follows:

First, note that "1 and 10" refers to a 1% management fee and a 10% incentive fee.

End-of-year capital = GBP140 million + GBP80 million = GBP220 million.

Management fee = GBP220 million × 1% = GBP2.2 million.

Incentive fee = (GBP220 million − GBP200 million) × 10% = GBP2 million.

If, however, the incentive fee is calculated after deduction of management fees (instead of being calculated independently), then the incentive fee would become (GBP220 million – GBP200 million – GBP2.2 million) × 10% = GBP1.78 million

Investor net return would have become (GBP220 – GBP200 – GBP2.2 – GBP1.78)/GBP200 ≈ 8.0%.

4. IRR vs. MOIC

Match the advantages and disadvantages of IRR and MOIC as performance measures for long-lived alternative investments:

Return Metrics	Advantages/Disadvantages
1. IRR	A. ignores timing of cash flows.
	B. considers timing of cash flows.
	C. is more complicated to calculate.
2. MOIC	D. is easy to calculate and understand.
	E. requires assumptions on opportunity costs and reinvestment rates.
	F. is the preferred measure for long-lived alternative investments.
	G. is commonly used by private equity and real estate investors as a shortcut.

Hide Solution

Solution:

1. IRR (F) is the preferred measure for long-lived alternative investments, because it (B) considers timing of cash flows, but it (C) is more complicated to calculate and (E) requires assumptions on opportunity costs and reinvestment rates.
2. MOIC (G) is commonly used by private equity and real estate investors as a shortcut, because it (D) is easy to calculate and understand, but it (A) ignores timing of cash flows.

5. Either/or Fee Structure

A closed-end infrastructure fund with initial capital of €100 million has an either/or fee structure under which the GP can either charge a 1% management fee or accept a higher 25% incentive fee, whichever is higher. All fees are calculated based on end-of-period net asset value. Standard high-water mark provisions apply.

The fund returns for the first five years are as follows:

Year	NAV (€ millions)	Gross Return
1	98.00	-2%
2	93.10	-5%
3	108.00	16%
4	129.60	20%
5	176.26	36%

Please calculate the fees received by the GP for each of the five years.

Hide Solution

Solution:

Year	NAV (€ millions)	Gross Return	Management Fee (%)	Incentive Fee (%)	Total Fee (€ millions)
1	98.00	–2%	1%	—	0.98
2	93.10	–5%	1%	—	0.93
3	108.00	16%	—	25% (108 – 100) × 25%	2.00
4	129.60	20%	—	25% (129.6 – 108.0) × 25%	5.40
5	176.26	36%	—	25% (176.26 – 129.60) × 25%	11.67
				Cumulative fees	**20.98**

Note that because the fund made a loss in Year 1 and in Year 2, there is no incentive fee and the GP can receive only the 1% management fee. In Year 3, when the fund made a profit, the incentive fee can be charged on the gain above the last high-water mark, which was the initial €100 million in Year 0. Since the management fee for Year 3 would have been just €1.08 million (= €108 million × 1%), the GP will be better off receiving the incentive fee of €2 million. Likewise, for Years 4 and 5, the GP can choose to receive the incentive fee at 25% of the profits in those two years without earning any management fee.

6. Hedge Fund Indexes

A common problem with hedge fund indexes is the upward bias due to:

A. backfill bias only.
B. survivorship bias only.
C. both backfill and survivorship bias.

Hide Solution

Solution:

C is correct. Both backfill bias and survivorship bias are common in hedge fund indexes.

Survivorship bias refers to the selection bias in the index due to the exclusion of failed funds from a given benchmark. It leads to overly optimistic return expectations. Backfill bias refers to hedge funds including only the most successful funds in an index a few years after inception. The subsequent inclusion or “backfilling” of prior performance data on a selective basis serves to increase average reported returns. Both of these biases result in an upward bias in hedge fund indexes because they may reflect not actual average hedge fund performance but, rather, only the returns of those funds that initially performed best and/or have not failed.

OPTIONAL UNTESTED SEGMENT

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AB For anybody who is confused, just know that the person who wrote the solutions was just as confused and they posted the correct solutions on here https://www.cfainstitute.org/sites/default/files/-/media/documents/support/programs/cfa/2025-cfa-level-i-errata.pdf. this is where you will find the errata and will help in correcting the doubt that CFAI might be wrong in their explanation

Created 4 months ago by Anupam Bhagbut — 1 reply | Last Activity: 2 months ago

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AK In Question Set, Q5 - it is funny that it says "Please calculate the fees received by the GP for each of the five years.", as if the request is coming directly from client.

Created 3 months ago by Ashank Kumar — 1 reply | Last Activity: 3 months ago

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MR In Example 4, Question 2 and 3, is the HWM 124.16 or 122.7?

Created 6 months ago by Mitchell Robert Cook — 4 replies | Last Activity: 3 months ago

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GR For question set problem 1, I am not understanding the catch-up of $4 million. Is it because the GPs 20% carried interest on the initial $100 million is $20 million, and since the investors receive $16 million in preferred returns, the remaining $4 million is the GPs share?

Created 3 months ago by Ganesh Raj Kafle — 0 replies | Last Activity: 3 months ago

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AG In example 4 - question 3, do we account for the 6% hurdle rate(specified in ex.4 q1) to be applicable? If yes, will that hurdle be calculated with respect to the HWM or the opening NAV?

LB

Why in ex. 5 performance fees are computed on gross basis instead of net basis (over the high water mark)? Am I missing something?

Created 5 months ago by Lorenzo Berti — 2 replies | Last Activity: 4 months ago

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NA

QSET3 is confusing.

They say that the value of the 2 funds are : "net of their respective management and incentive fees" but they still deduct them later.

Created 4 months ago by Nada Ait labsir — 1 reply | Last Activity: 4 months ago

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PC

In Ex. 4 Question 2, The High Water Mark hasn't been computed net of fees. It should be P2 * rm + {max [0, (P2 (1 - rm) - PHWM)] * p}. Can anyone please explain whether this understanding is correct?

Created 5 months ago by Priyesh Choodamani Iyer — 1 reply | Last Activity: 4 months ago

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MP

isnt it that for soft hurdle the perf fees should be computed as per the total 160 -4 = 56 (therefore return for GP is 11.2, plus catch up, 15.2). For me the CFA program is caclcualting this as a hard hurdle rate.

Created 5 months ago by Manuel Paredes Caldito — 0 replies | Last Activity: 5 months ago

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PZ

can anyone explain the answer to Question2 in Example 7 "*2. Calculate the return to the investor from investing in XYZ FOF. Assume that the other investments in the XYZ FOF portfolio generate the same return before management fees as those of ABC HF.*"

Why is the return 14% or 14m?

Created 6 months ago by PIERFRANCESCO ZEOLI — 1 reply | Last Activity: 6 months ago

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